

Fourier Transform properties

Linearity, shift, symmetry, convolution, windows, FFT and analysis/synthesis

Digital Signal Processing for Sound and Music · Topic 3

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TOPIC 3

Lecture roadmap

- | | | | |
|----------|--|----------|--|
| 1 | Linearity, shift and symmetry | 2 | Convolution and filtering intuition |
| 3 | Energy, dB and phase unwrapping | 4 | Zero-padding and zero-phase FFT buffers |
| 5 | FFT and analysis/synthesis workflow | 6 | Example notebook + Exercise 3 |

Linearity:

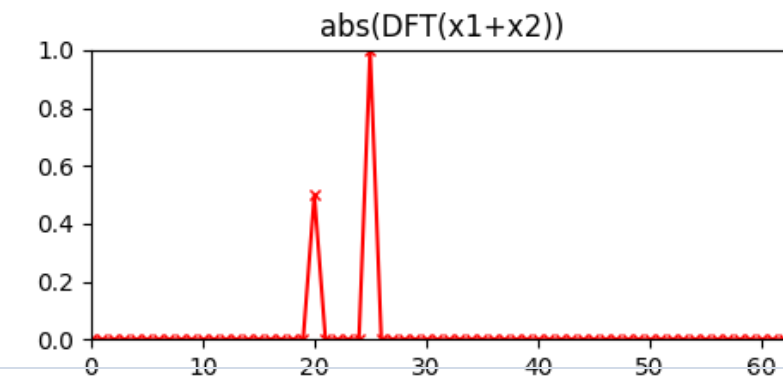
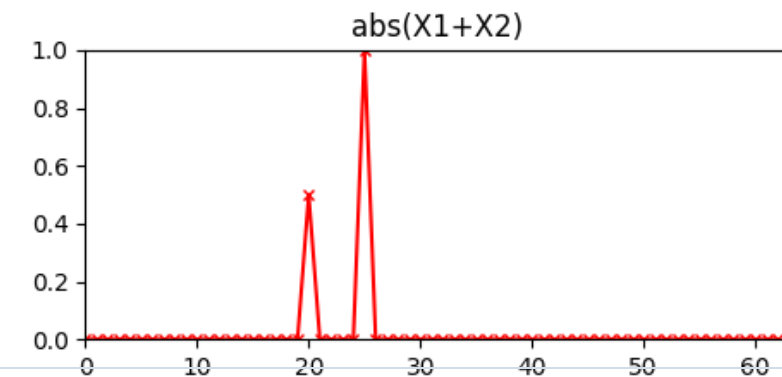
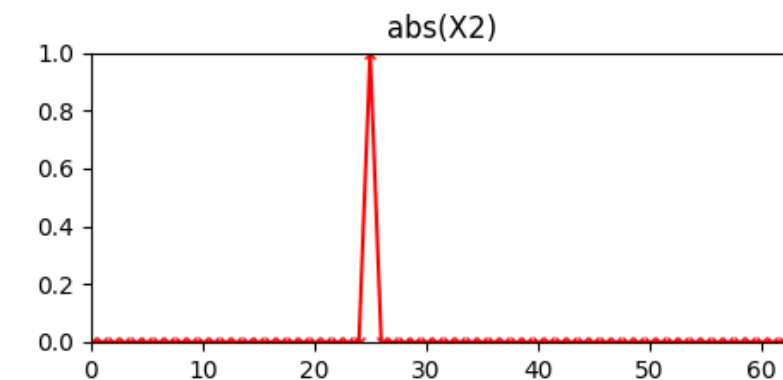
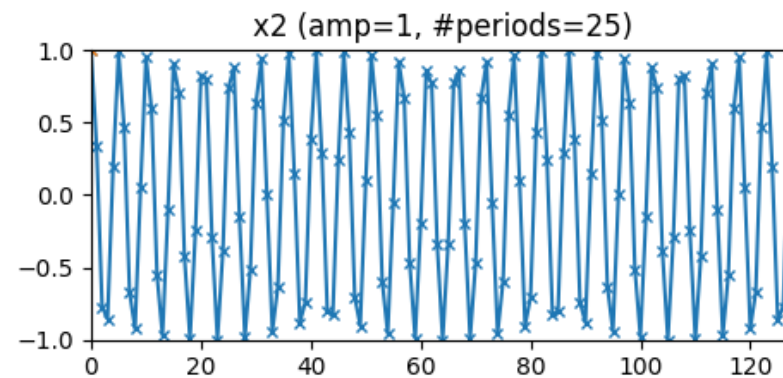
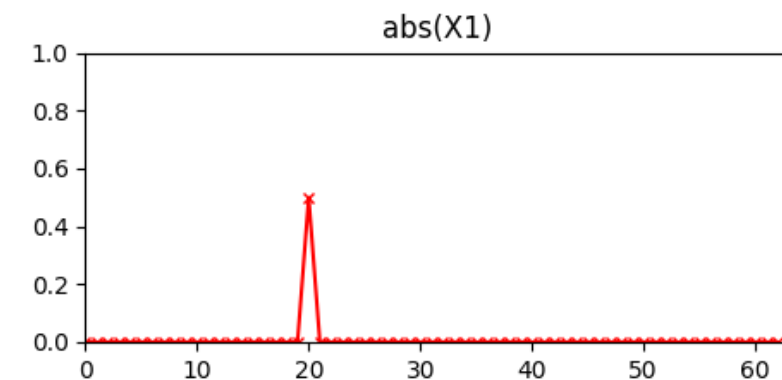
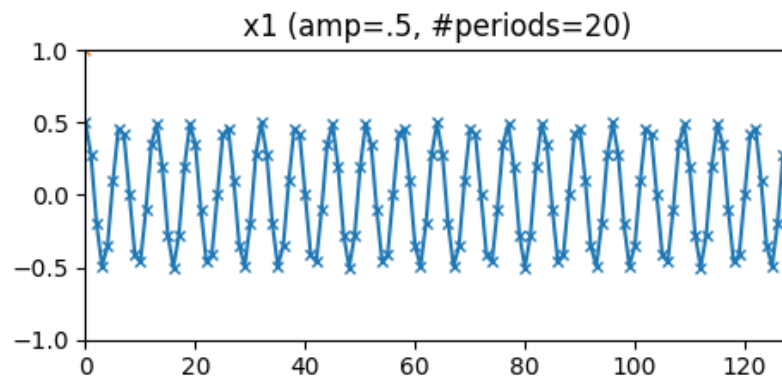
Property: $ax_1[n] + bx_2[n] \Leftrightarrow aX_1[k] + bX_2[k]$

$$\begin{aligned}
 & DFT(ax_1[n] + bx_2[n]) \\
 &= \sum_{n=0}^{N-1} (ax_1[n] + bx_2[n])e^{-j2\pi kn/N} \\
 &= \sum_{n=0}^{N-1} a x_1[n]e^{-j2\pi kn/N} + \sum_{n=0}^{N-1} b x_2[n]e^{-j2\pi kn/N} \\
 &= a \sum_{n=0}^{N-1} x_1[n]e^{-j2\pi kn/N} + b \sum_{n=0}^{N-1} x_2[n]e^{-j2\pi kn/N} \\
 &= aX_1[k] + bX_2[k]
 \end{aligned}$$

Linearity

- DFT is a linear transform.
- Analyze sums by summing spectra.
- Useful for additive sound models.

Linearity: spectra add



Shift:

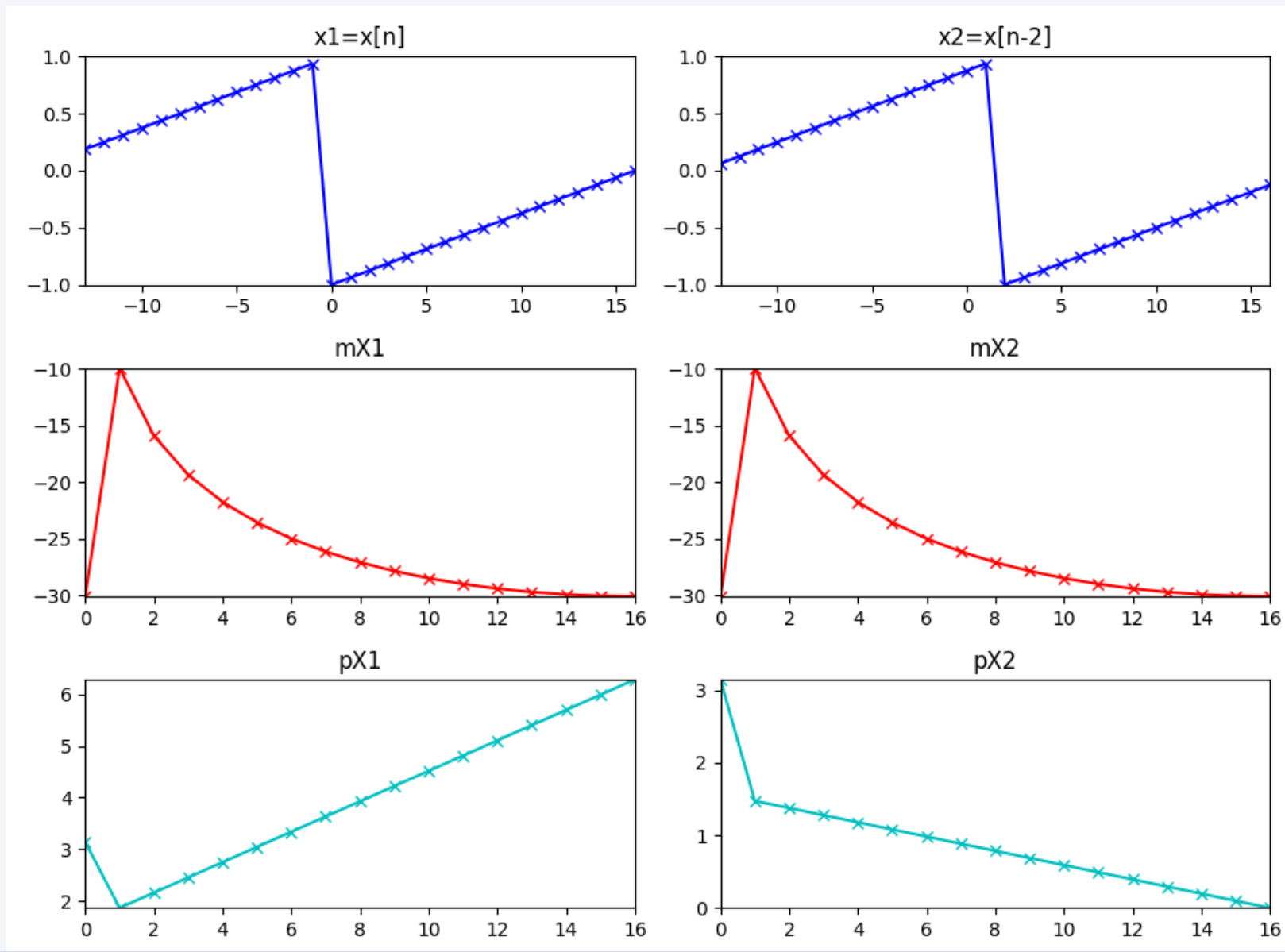
Property: $x[n - n_0] \Leftrightarrow e^{-j2\pi kn_0/N} X[k]$

$$\begin{aligned}
 & DFT(x[n - n_0]) \\
 &= \sum_{n=0}^{N-1} x[n - n_0] e^{-j2\pi kn/N} \\
 &= \sum_{m=-n_0}^{N-1-n_0} x[m] e^{-j2\pi k(m+n_0)/N} \quad (m = n - n_0) \\
 &= \sum_{m=0}^{N-1} x[m] e^{-j2\pi km/N} e^{-j2\pi kn_0/N} \\
 &= e^{-j2\pi kn_0/N} \sum_{m=0}^{N-1} x[m] e^{-j2\pi km/N} \\
 &= e^{-j2\pi kn_0/N} X[k]
 \end{aligned}$$

Shift

- A time shift does not change magnitude.
- It adds a linear phase term.
- This is why centering matters.

Shift: same magnitude, phase rotation

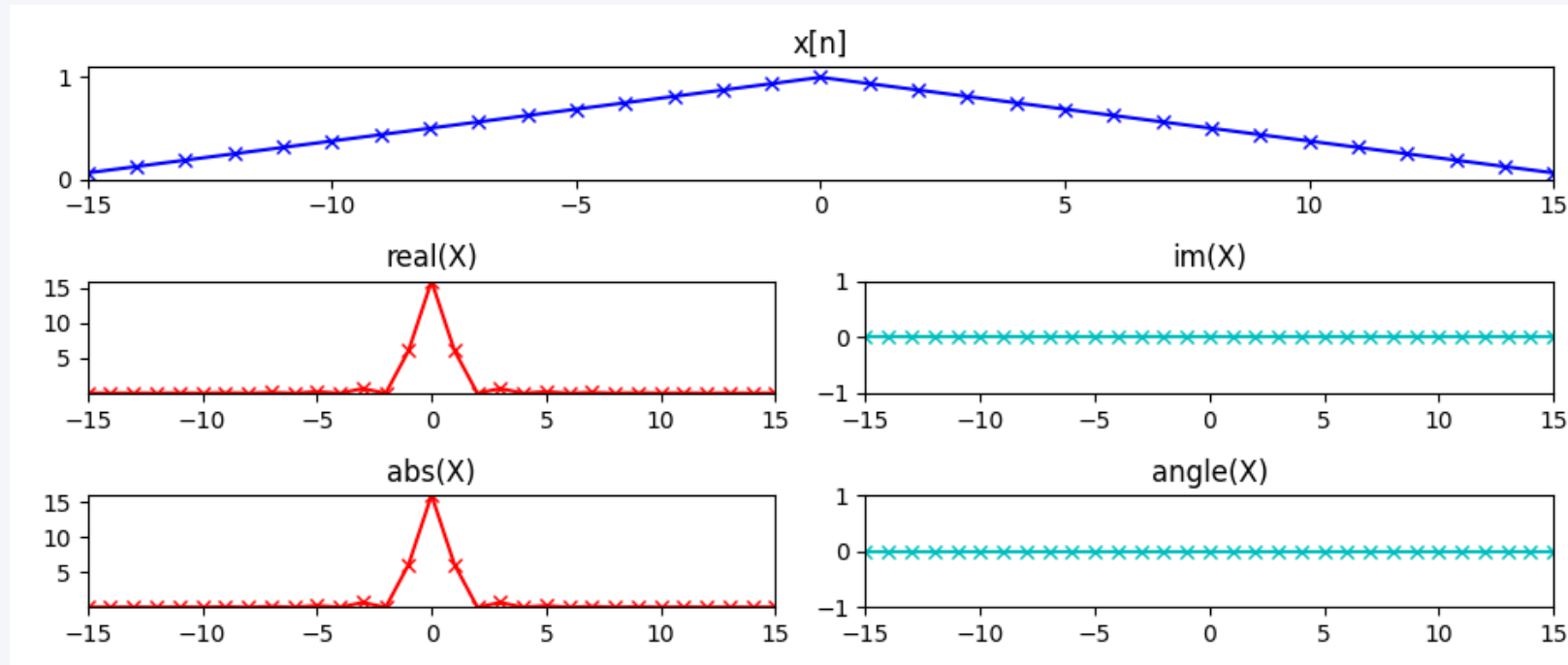


Symmetry:

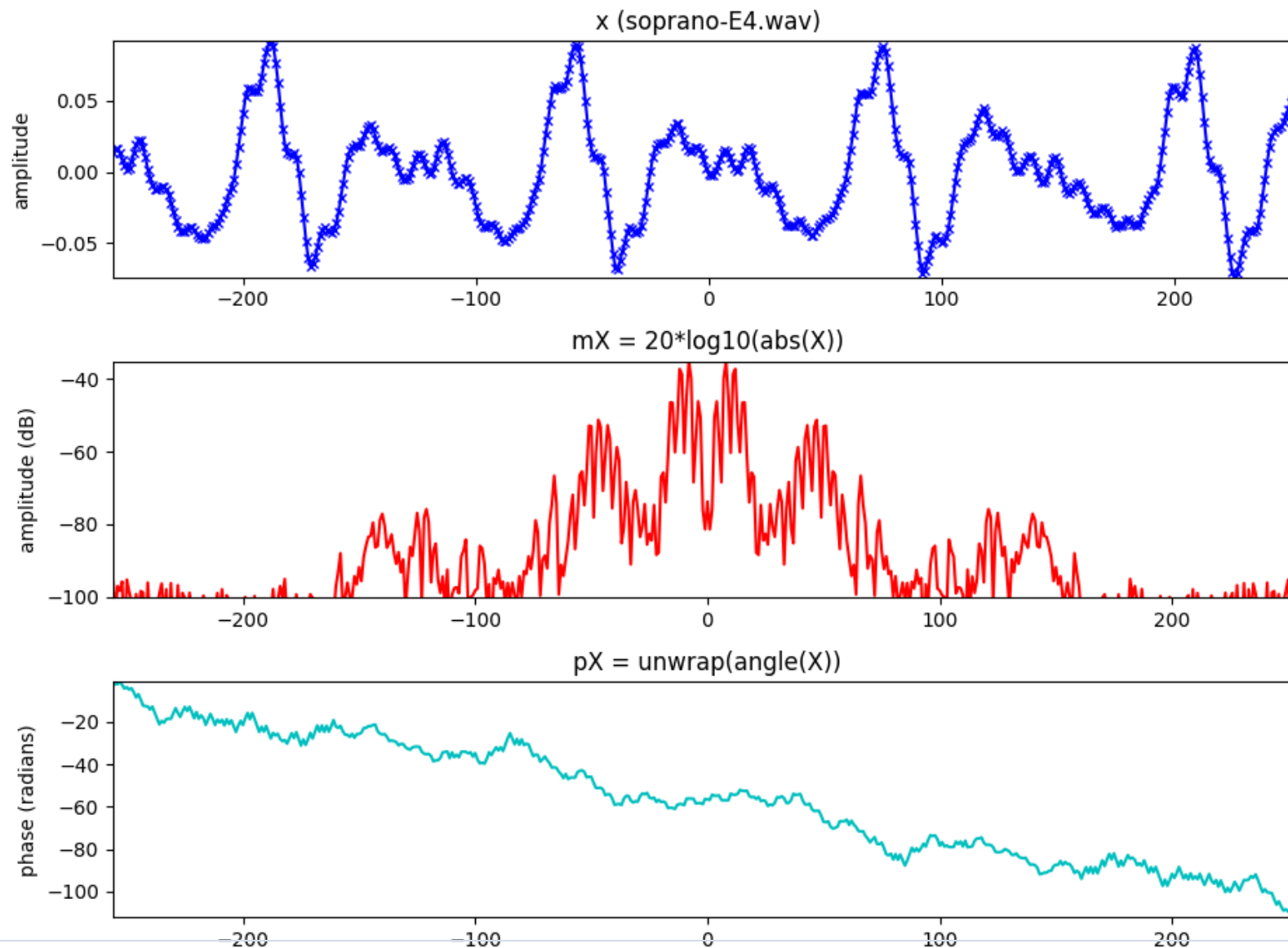
$$\begin{aligned}
 x[n] \text{ real} &\Leftrightarrow \Re\{X[k]\} \text{ even and } \Im\{X[k]\} \text{ odd} \\
 &\Leftrightarrow |X[k]| \text{ even and } \angle X[k] \text{ odd} \\
 x[n] \text{ real and even} &\Leftrightarrow \Re\{X[k]\} \text{ even and } \Im\{X[k]\} = 0 \\
 &\Leftrightarrow |X[k]| \text{ even and } \angle X[k] = n\pi
 \end{aligned}$$

Symmetry

- Real signals have conjugate-symmetric spectra.
- Real + even signals have real spectra.
- These properties reduce storage and computation.



Symmetry for a centered real signal



Convolution:

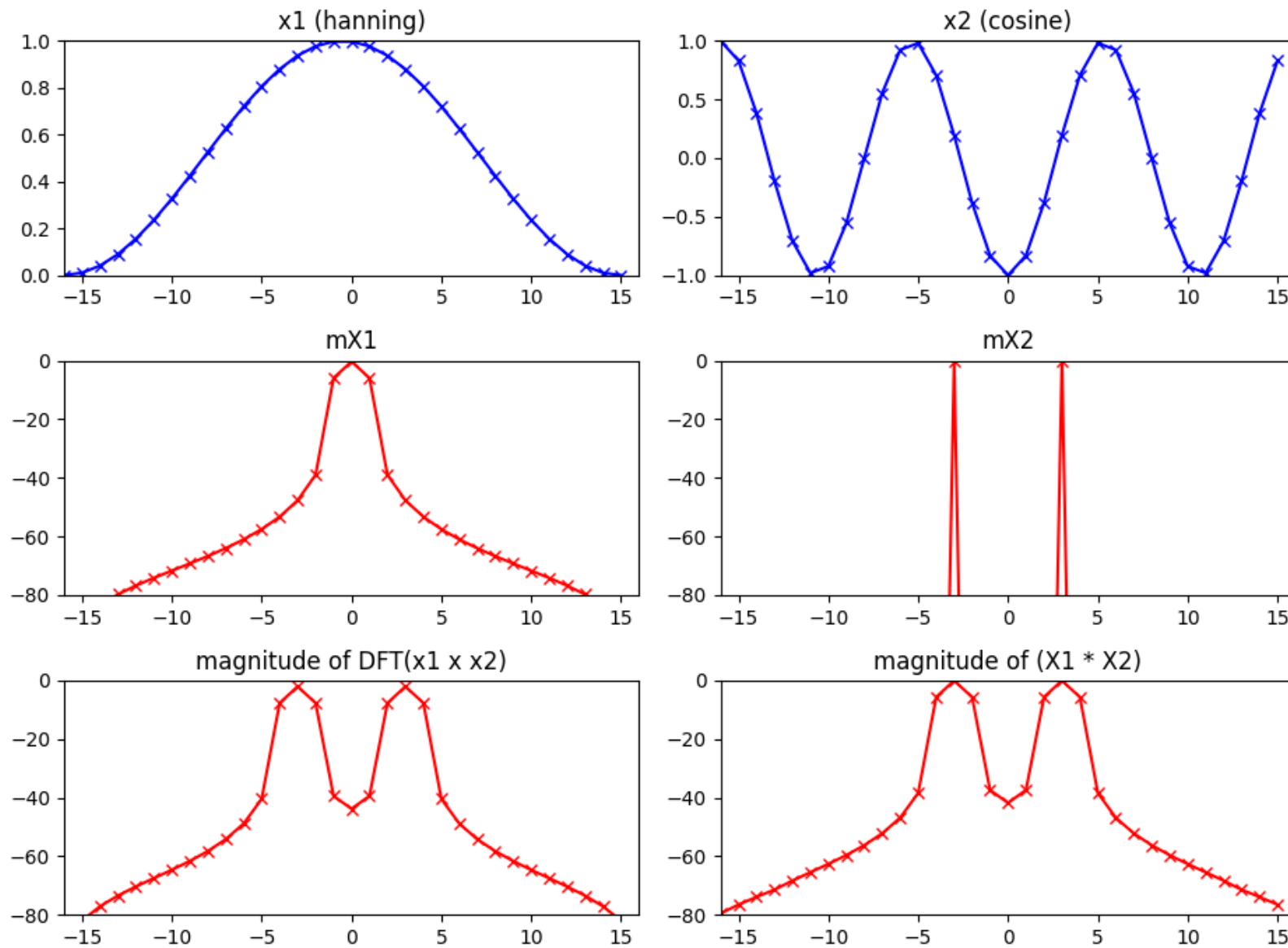
Property: $x_1[n] * x_2[n] \Leftrightarrow X_1[k] \times X_2[k]$

$$\begin{aligned}
 & DFT(x_1[n] * x_2[n]) \\
 &= \sum_{n=0}^{N-1} (x_1[n] * x_2[n]) e^{-j2\pi kn/N} \\
 &= \sum_{n=0}^{N-1} \sum_{m=0}^{N-1} x_1[m] x_2[n-m] e^{-j2\pi kn/N} \\
 &= \sum_{m=0}^{N-1} x_1[m] \sum_{n=0}^{N-1} x_2[n-m] e^{-j2\pi kn/N} \\
 &= \left(\sum_{m=0}^{N-1} x_1[m] e^{-j2\pi km/N} \right) X_2[k] \\
 &= X_1[k] \times X_2[k]
 \end{aligned}$$

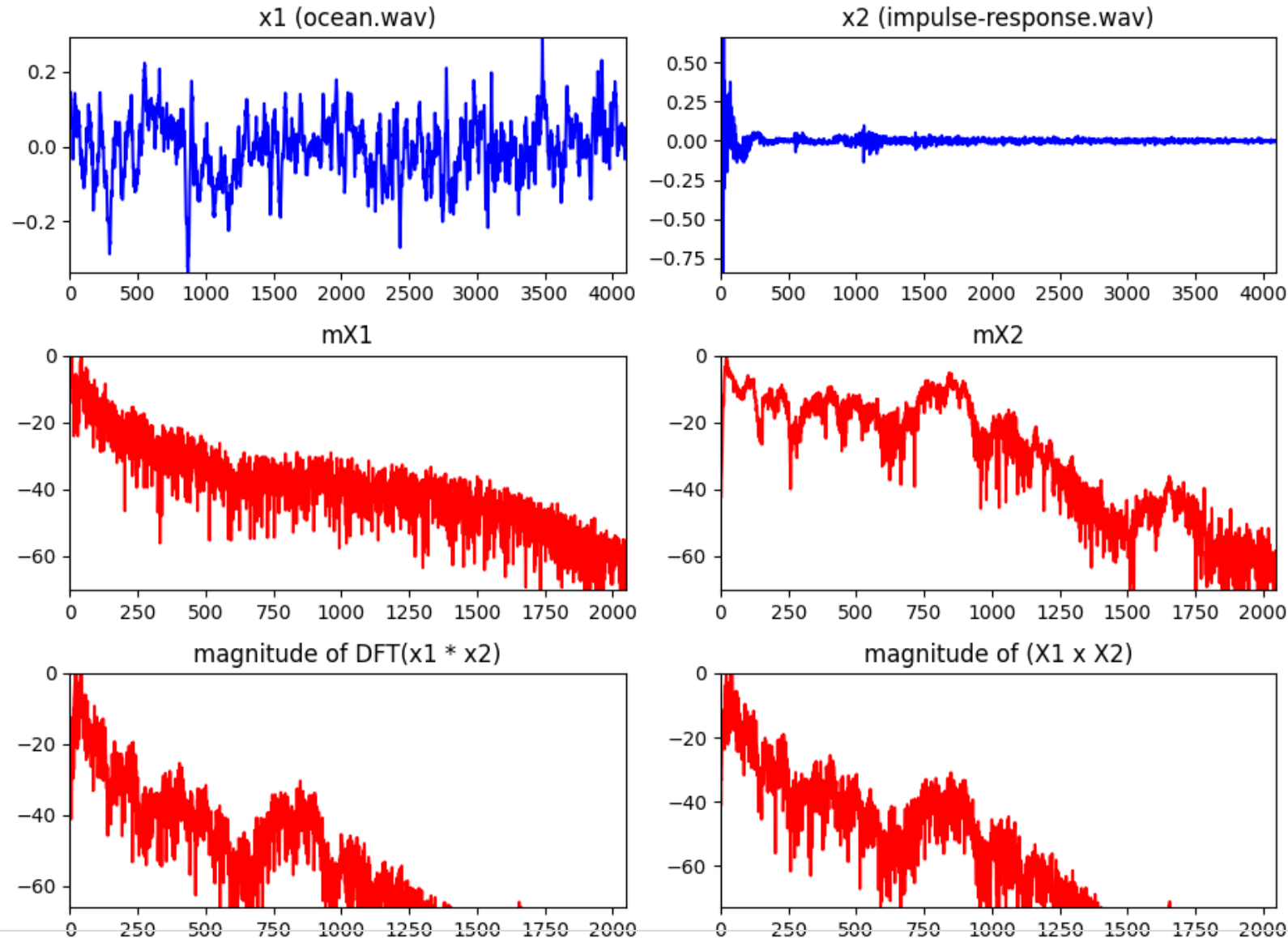
Property:

- Convolution in time becomes multiplication in frequency.
- This is the basis of filtering.
- Circular convolution follows directly from the DFT.

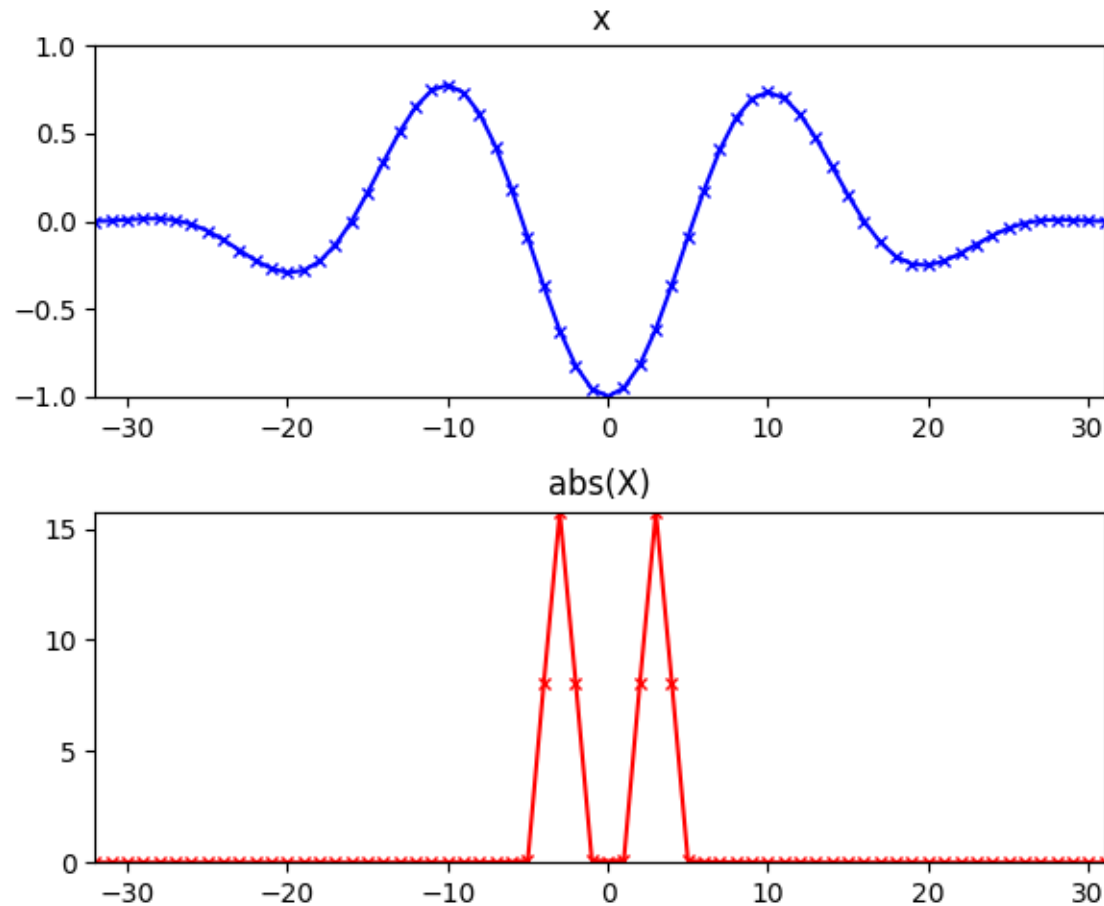
Convolution property



Convolution as frequency-domain filtering



Energy conservation



Parseval identity:

$$\sum |x[n]|^2 = (1/N) \sum |X[k]|^2$$

time energy:

$$\sum |x[n]|^2 = 11.81182$$

frequency energy:

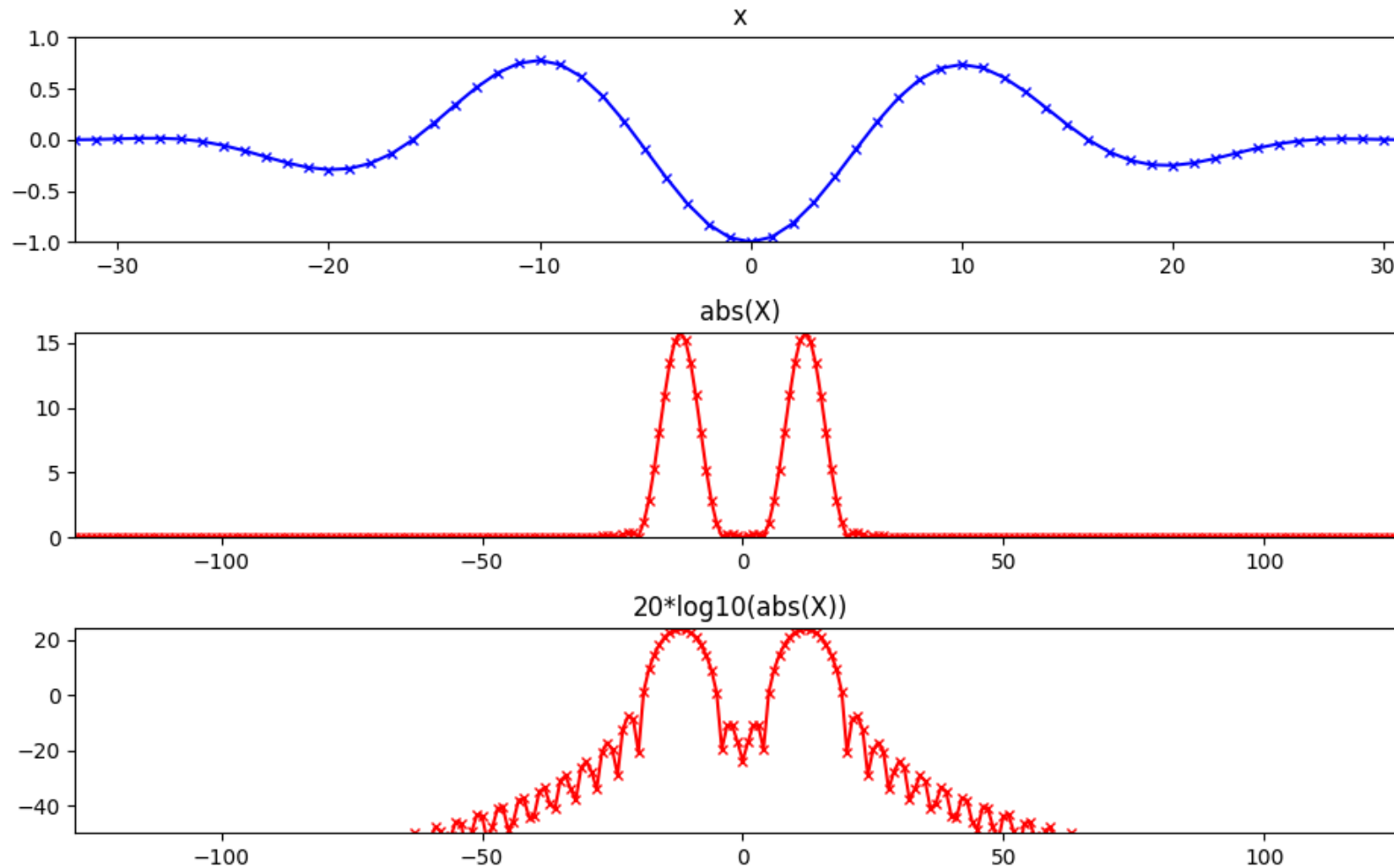
$$(1/N) \sum |X[k]|^2 = 11.81182$$

Parseval

- Energy is conserved across domains.
- Frequency-domain energy has $1/N$ scaling.
- Useful for checking implementations.

Amplitude in decibels (dB)

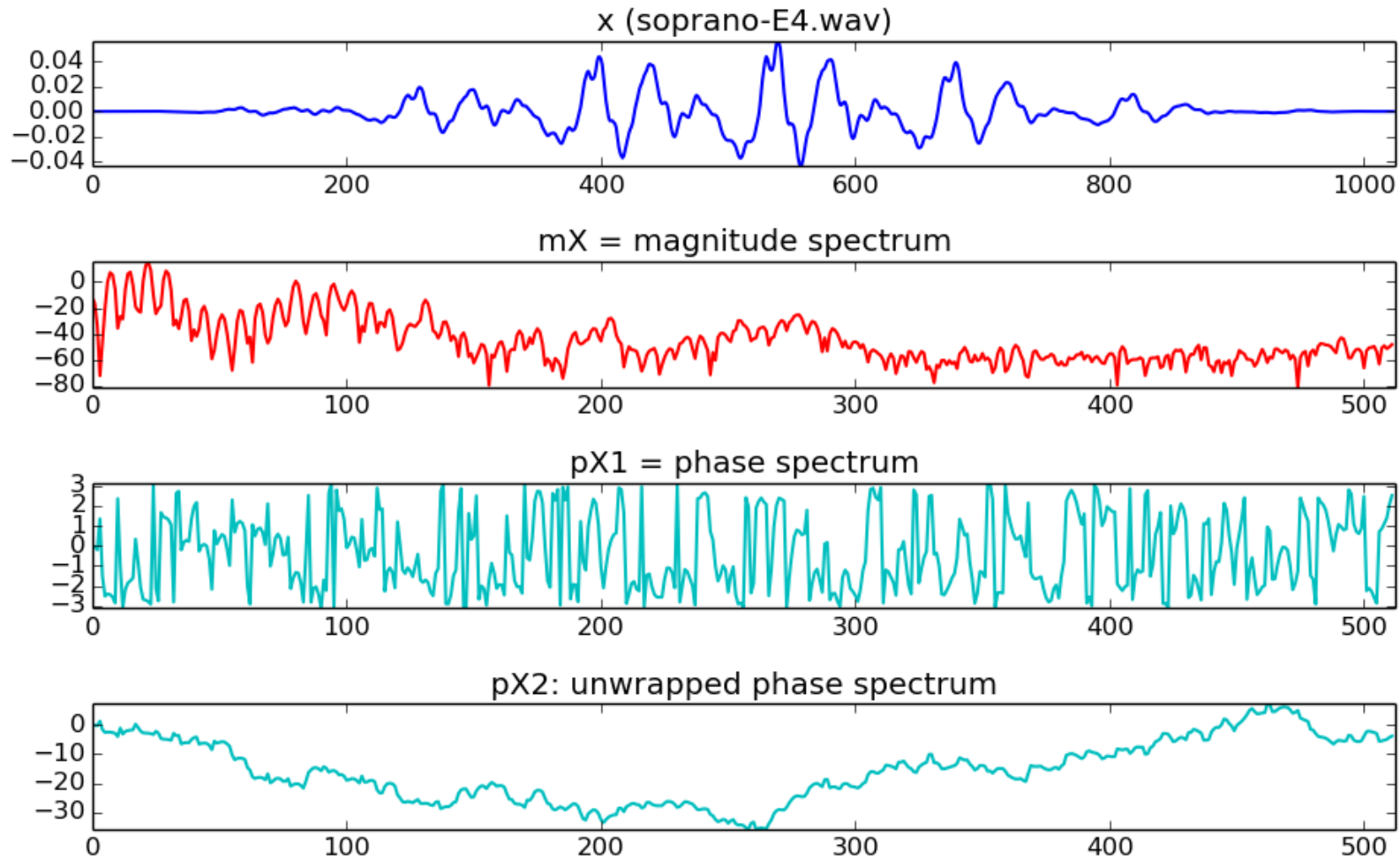
$$X_{\text{dB}}[k] = 20 \log_{10}(\max(|X[k]|, \epsilon) / \text{ref})$$



dB scale

- $20 \log_{10}$ maps amplitude ratios to dB.
- Small magnitudes become visible.
- Use a floor before log to avoid $-\infty$.

Phase unwrapping



Phase

- Wrapped phase jumps at $\pm\pi$.
- Unwrapping reveals continuous trends.
- Shifts produce phase ramps.

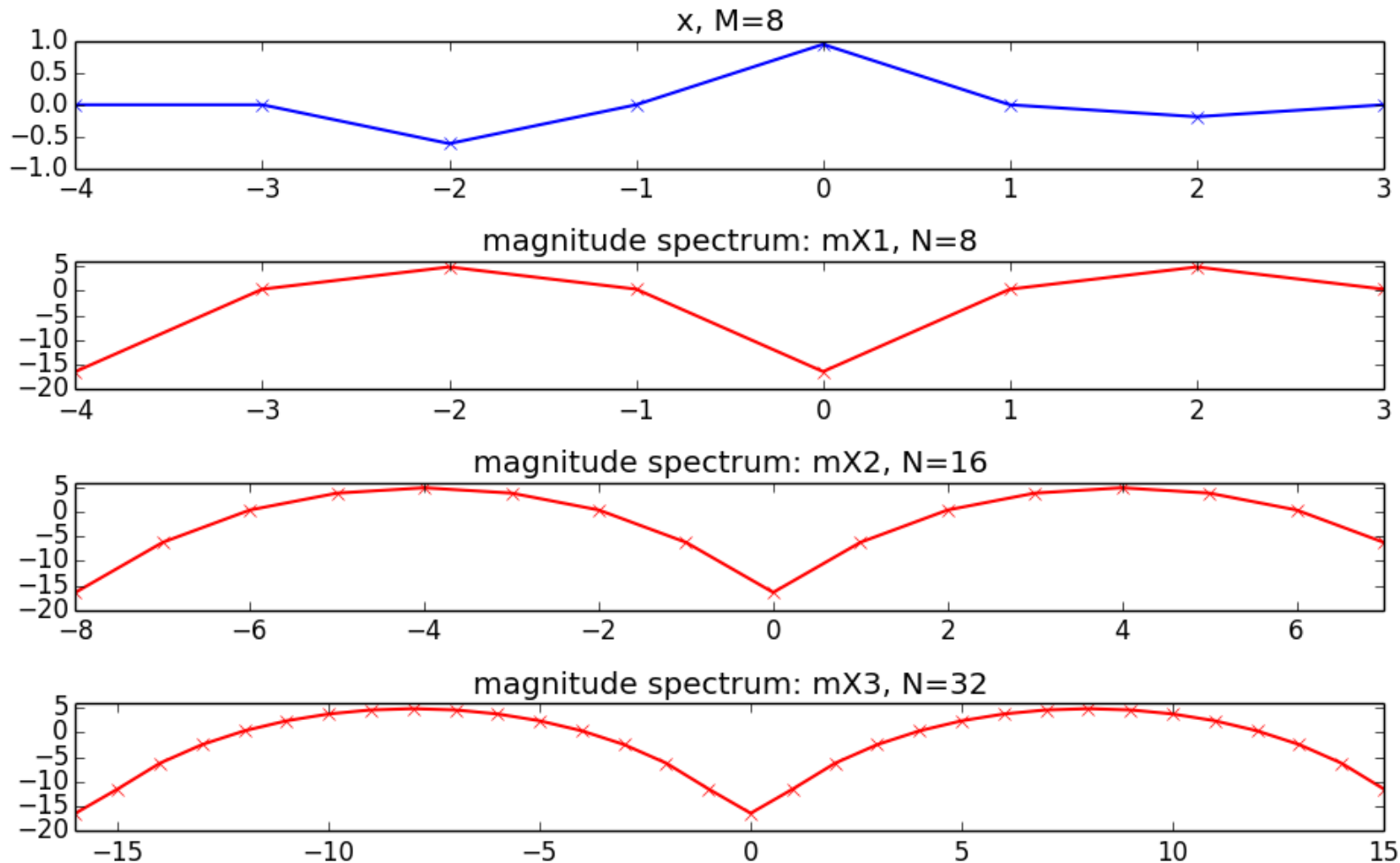
Zero-padding

zero-padding $\Leftrightarrow$ interpolation

$$x_{zp}[n] = x[n] \text{ for } 0 \leq n < M; 0 \text{ otherwise}$$

Zero-padding

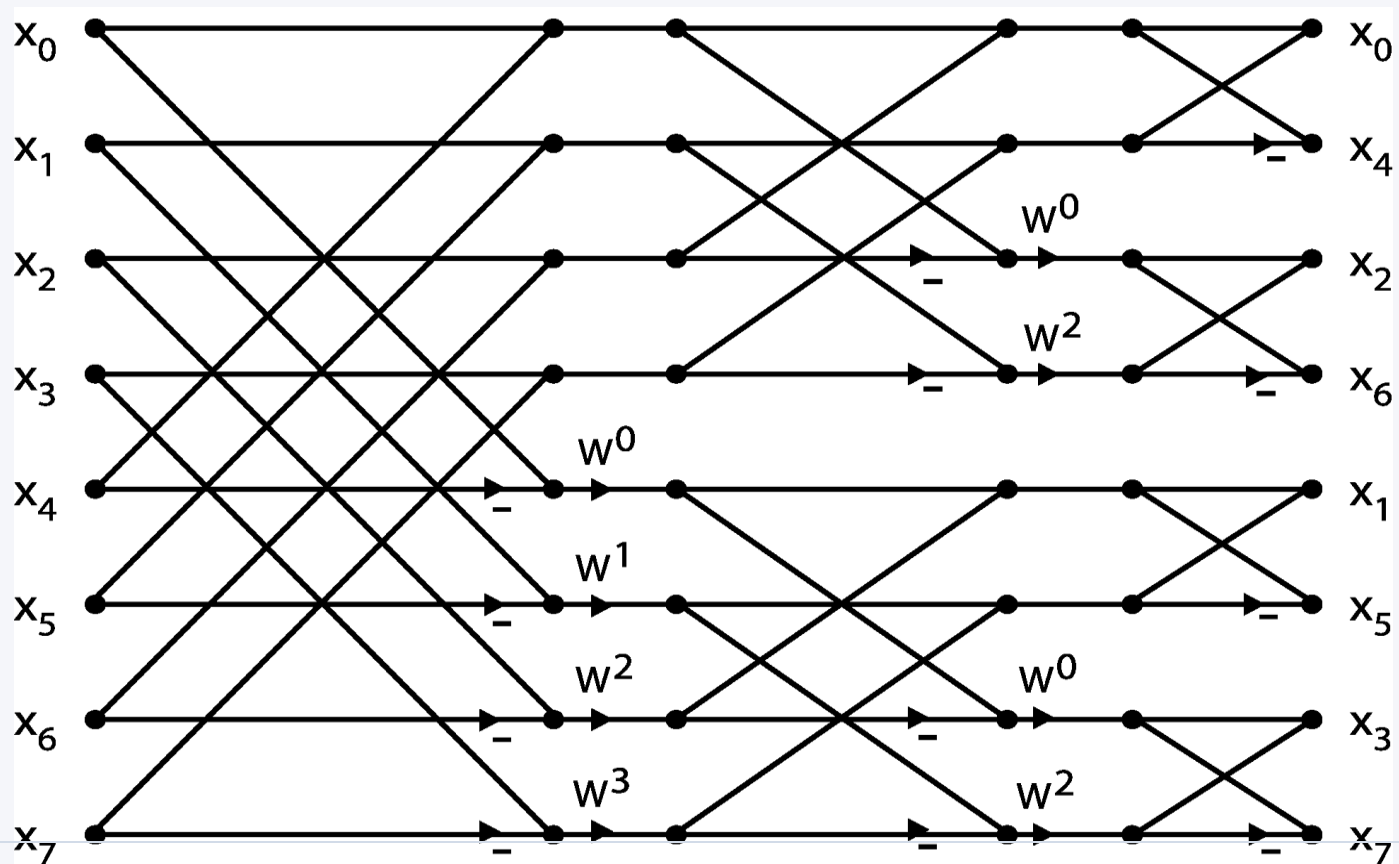
- Does not add information.
- Interpolates the DFT samples.
- Helps visualize and locate peaks.



Fast Fourier Transform

Cooley-Tukey algorithm: breaks down recursively the DFT of a power of 2 size into two pieces of size $N/2$.

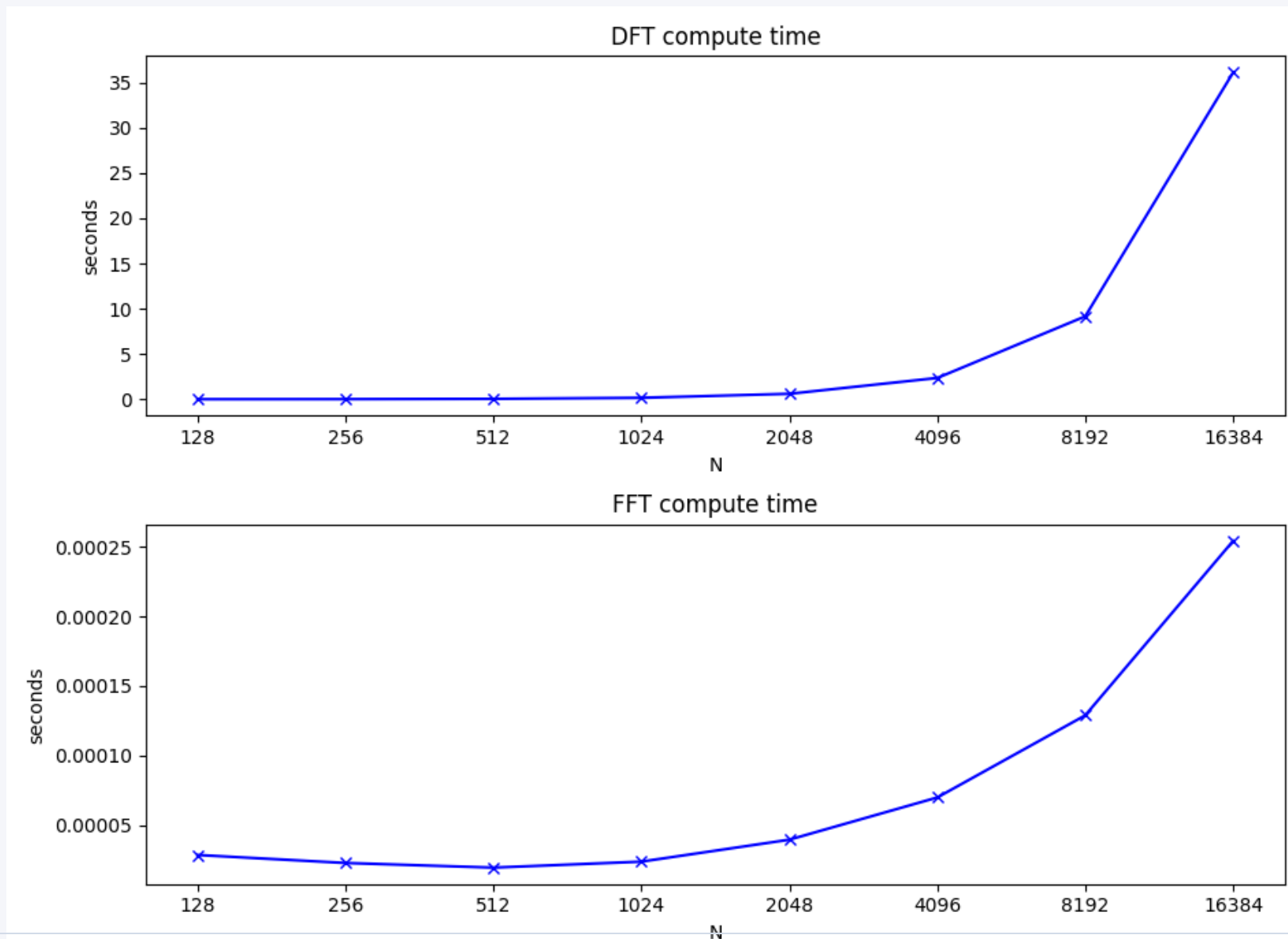
DFT: $O(N^2)$ FFT: $O(N \log_2 N)$



FFT

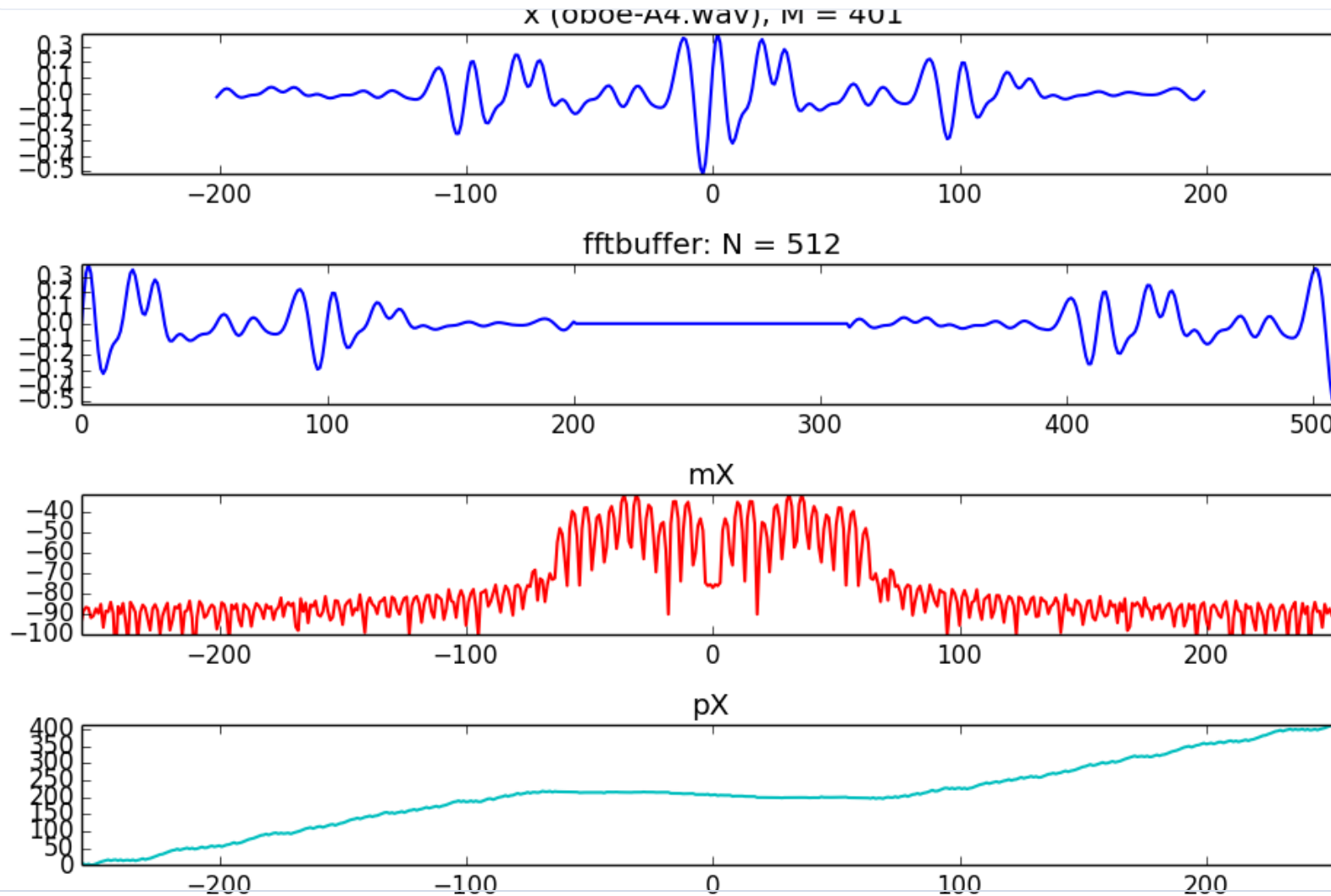
- FFT computes the same DFT.
- It reduces cost from $O(N^2)$ to $O(N \log N)$.
- Power-of-two sizes are often efficient.

DFT vs FFT computational cost



FFT and zero-phase windowing

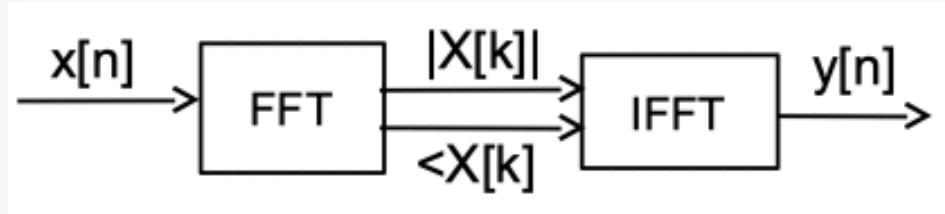
zero-phase FFT: center the frame around $n=0$ before computing $X[k]$



Zero phase

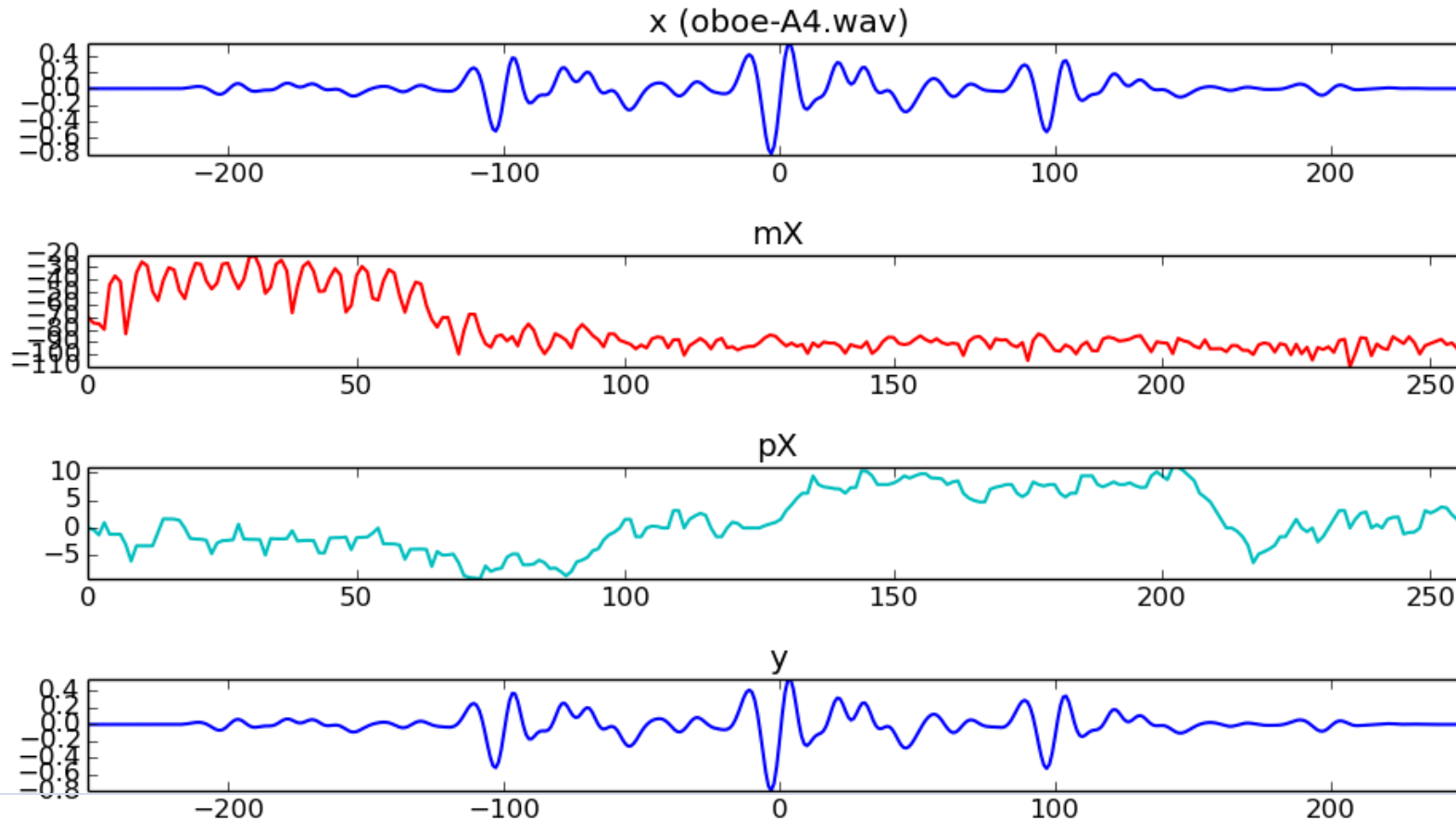
- Center the frame around $n = 0$.
- Reorder samples into the FFT buffer.
- Avoid an artificial phase ramp.

Analysis/synthesis with magnitude and phase



Analysis / synthesis

- FFT decomposes the frame.
- Magnitude and phase preserve information.
- IFFT reconstructs the signal.



Example notebooks: Fourier properties in practice

Use `examples/dft-examples.ipynb` to verify that every spectrum is tied to a precise time-domain operation: framing, windowing, centering, FFT, and inverse FFT.

1 • Frame and center

Select one short sound frame and place it around $n = 0$ when phase interpretation matters. A non-centered frame introduces a phase ramp.

2 • Window the frame

Apply a Hann/Hamming-type window to reduce boundary discontinuities. Then the spectral shape belongs to the windowed signal, not the raw frame.

3 • Inspect spectrum

Look at magnitude and phase together. For real signals, use conjugate symmetry as a consistency check: magnitudes mirror, phases oppose.

4 • Reconstruct

Use the full complex spectrum for IFFT. Zero-padding can make plots smoother, but it does not add information.

Key reading strategy: whenever a plot is shown, ask which exact time-domain frame, window, centering convention, and FFT size produced that spectrum.

How to read the example outputs

Windowing

Compare the spectrum before and after windowing. A smoother window reduces discontinuities at the frame boundaries and controls spectral leakage.

Symmetry

For real-valued frames, check that positive and negative frequency bins are conjugate pairs. Magnitudes mirror; phases have opposite signs.

Zero-phase / phase

Centering the frame around $n = 0$ removes the artificial phase ramp caused by a time shift. This makes phase easier to interpret.

Use the example as a bridge from the equations to Exercise 3: every visual property should correspond to a testable relation in Python.

What to remember from Topic 3

- Fourier properties explain how time-domain operations affect spectra.
- Centering and windowing are essential for interpretable phase.
- Zero-padding interpolates; it does not increase resolution.
- FFT is an efficient DFT implementation.
- Analysis/synthesis requires both magnitude and phase.

Exercise 3 — Fourier properties

Notebook: [exercises/E3-Fourier-properties.ipynb](#)

1 • Leakage and window length

Choose a frame size M so sinusoids fit an integer number of periods when possible. Compare with a mismatched length to see leakage.

2 • Symmetry tests

Use real, even, and odd signals. Check conjugate symmetry, real spectra, and imaginary spectra after zero-phase placement.

3 • Frequency-domain filtering

Suppress or keep selected spectral components, then synthesize with the inverse DFT and listen/inspect the result.

4 • FFT size and zero-padding

Compare window size and FFT size. Zero-padding interpolates the spectrum; it does not add new information.

What to submit/check: clear Python functions, plots of magnitude and phase, and short comments explaining whether the numerical result matches the theoretical property.

Goal: turn each Fourier property into a small experiment with a measurable numerical check and a plot that confirms the expected behaviour.

What you will implement in Exercise 3

- Choose M to minimize spectral leakage for two sinusoids using the LCM of periods.
- Test real/even/odd symmetry using zero-phase windowing and inspect real/imaginary parts.
- Manipulate spectral components in the DFT domain and synthesize the filtered signal.
- Compare window size M and FFT size N to separate resolution from zero-padding interpolation.
- Always verify with both magnitude and phase, not magnitude alone.

References and credits

- Original MTG sms-tools-materials lecture slides and plotting code.
- plots-code: [lectures/03-Fourier-properties/plots-code](#)
- Example notebook: [examples/dft-examples.ipynb](#)
- Exercise notebook: [exercises/E3-Fourier-properties.ipynb](#)
- Julius O. Smith, Mathematics of the Discrete Fourier Transform.
- Slides follow the CC Attribution-NonCommercial-ShareAlike spirit of the original materials.